

Registry

Search all resources

Use Terraform Cloud for free



vpc
AWS

Version 5.1.2 (latest)

Terraform module which creates VPC resources on AWS ua

Published September 7, 2023 by terraform-aws-modules
Module managed by antonbabenko
Source Code: github.com/terraform-aws-modules/terraform-aws-vpc (report an issue)

Submodules

Examples

Module Downloads

Downloads this week

Downloads this month

Downloads this year

Downloads over all time

Provision Instructions

Copy and paste into your Terraform configuration, insert the variables

```
module "vpc" {  
  source = "terraform-aws-modules/vpc/aws"  
  version = "5.1.2"  
}
```

Readme

Inputs (220)

Outputs (109)

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our Privacy Policy and Cookie Policy.

Manage Preferences

Dismiss

[Dependency \(1\)](#)[Resources \(80\)](#)

AWS VPC Terraform module

Terraform module which creates VPC resources on AWS.

Russia invaded Ukraine, killing tens of thousands of civilians and displacing millions more. It's a genocide. Please help us defend freedom, democracy and Ukraine's right to exist.

[Help Ukraine Now →](#)

Usage

```
module "vpc" {  
  source = "terraform-aws-modules/vpc/aws"  
  
  name = "my-vpc"  
  cidr = "10.0.0.0/16"  
  
  azs          = ["eu-west-1a", "eu-west-1b", "eu-west-1c"]  
  private_subnets = ["10.0.1.0/24", "10.0.2.0/24", "10.0.3.0/24"]  
  public_subnets  = ["10.0.101.0/24", "10.0.102.0/24", "10.0.103.0/24"]  
  
  enable_nat_gateway = true
```

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

```
Terraform = "true"
Environment = "dev"
}
}
```

External NAT Gateway IPs

By default this module will provision new Elastic IPs for the VPC's NAT Gateways. This means that when creating a new VPC, new IPs are allocated, and when that VPC is destroyed those IPs are released. Sometimes it is handy to keep the same IPs even after the VPC is destroyed and re-created. To that end, it is possible to assign existing IPs to the NAT Gateways. This prevents the destruction of the VPC from releasing those IPs, while making it possible that a re-created VPC uses the same IPs.

To achieve this, allocate the IPs outside the VPC module declaration.

```
resource "aws_eip" "nat" {
  count = 3

  vpc = true
}
```

Then, pass the allocated IPs as a parameter to this module.

```
module "vpc" {
  source = "terraform-aws-modules/vpc/aws"

  # The rest of arguments are omitted for brevity

  enable_nat_gateway = true
  single_nat_gateway = false
  reuse_nat_ips      = true          # <= Skip creation of EIPs for the NAT Gateways
  external_nat_ip_ids = "${aws_eip.nat.*.id}" # <= IPs specified here as input to the module
}
```

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

1 IP. Passing the IPs into the module is done by setting two variables `reuse_nat_ips = true` and `external_nat_ip_ids = "${aws_eip.nat.*.id}"`.

NAT Gateway Scenarios

This module supports three scenarios for creating NAT gateways. Each will be explained in further detail in the corresponding sections.

- One NAT Gateway per subnet (default behavior)

- `enable_nat_gateway = true`
- `single_nat_gateway = false`
- `one_nat_gateway_per_az = false`

- Single NAT Gateway

- `enable_nat_gateway = true`
- `single_nat_gateway = true`
- `one_nat_gateway_per_az = false`

- One NAT Gateway per availability zone

- `enable_nat_gateway = true`
- `single_nat_gateway = false`
- `one_nat_gateway_per_az = true`

If both `single_nat_gateway` and `one_nat_gateway_per_az` are set to `true`, then `single_nat_gateway` takes precedence.

One NAT Gateway per subnet (default)

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

number of `intra_subnets`, since the latter are designed to have no Internet access via NAT Gateway. For example, if your configuration looks like the following:

```
database_subnets    = ["10.0.21.0/24", "10.0.22.0/24"]
elasticache_subnets = ["10.0.31.0/24", "10.0.32.0/24"]
private_subnets     = ["10.0.1.0/24", "10.0.2.0/24", "10.0.3.0/24", "10.0.4.0/24", "10.0.5.0/24"]
redshift_subnets    = ["10.0.41.0/24", "10.0.42.0/24"]
intra_subnets       = ["10.0.51.0/24", "10.0.52.0/24", "10.0.53.0/24"]
```

Then `5` NAT Gateways will be created since `5` private subnet CIDR blocks were specified.

Single NAT Gateway

If `single_nat_gateway = true`, then all private subnets will route their Internet traffic through this single NAT gateway. The NAT gateway will be placed in the first public subnet in your `public_subnets` block.

One NAT Gateway per availability zone

If `one_nat_gateway_per_az = true` and `single_nat_gateway = false`, then the module will place one NAT gateway in each availability zone you specify in `var.azs`. There are some requirements around using this feature flag:

- The variable `var.azs` **must** be specified.
- The number of public subnet CIDR blocks specified in `public_subnets` **must** be greater than or equal to the number of availability zones specified in `var.azs`. This is to ensure that each NAT Gateway has a dedicated public subnet to deploy to.

"private" versus "intra" subnets

By default, if NAT Gateways are enabled, private subnets will be configured with routes for Internet traffic that point at the NAT Gateways configured by use of the above options.

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

functions only need to pass traffic to internal resources or VPC endpoints for AWS services.

Since AWS Lambda functions allocate Elastic Network Interfaces in proportion to the traffic received ([read more](#)), it can be useful to allocate a large private subnet for such allocations, while keeping the traffic they generate entirely internal to the VPC.

You can add additional tags with `intra_subnet_tags` as with other subnet types.

VPC Flow Log

VPC Flow Log allows to capture IP traffic for a specific network interface (ENI), subnet, or entire VPC. This module supports enabling or disabling VPC Flow Logs for entire VPC. If you need to have VPC Flow Logs for subnet or ENI, you have to manage it outside of this module with [aws_flow_log resource](#).

VPC Flow Log Examples

By default `file_format` is `plain-text`. You can also specify `parquet` to have logs written in Apache Parquet format.

```
flow_log_file_format = "parquet"
```

Permissions Boundary

If your organization requires a permissions boundary to be attached to the VPC Flow Log role, make sure that you specify an ARN of the permissions boundary policy as `vpc_flow_log_permissions_boundary` argument. Read more about required [IAM policy for publishing flow logs](#).

Conditional creation

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

```
# This VPC will not be created
module "vpc" {
  source = "terraform-aws-modules/vpc/aws"

  create_vpc = false
  # ... omitted
}
```

Public access to RDS instances

Sometimes it is handy to have public access to RDS instances (it is not recommended for production) by specifying these arguments:

```
create_database_subnet_group      = true
create_database_subnet_route_table = true
create_database_internet_gateway_route = true

enable_dns_hostnames = true
enable_dns_support   = true
```

Network Access Control Lists (ACL or NACL)

This module can manage network ACL and rules. Once VPC is created, AWS creates the default network ACL, which can be controlled using this module (`manage_default_network_acl = true`).

Also, each type of subnet may have its own network ACL with custom rules per subnet. Eg, set `public_dedicated_network_acl = true` to use dedicated network ACL for the public subnets; set values of `public_inbound_acl_rules` and `public_outbound_acl_rules` to specify all the NACL rules you need to have on public subnets (see `variables.tf` for default values and structures).

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Public access to Redshift cluster

Sometimes it is handy to have public access to Redshift clusters (for example if you need to access it by Kinesis - VPC endpoint for Kinesis is not yet supported by Redshift) by specifying these arguments:

```
enable_public_redshift = true # <= By default Redshift subnets will be associated with the private route table
```

Transit Gateway (TGW) integration

It is possible to integrate this VPC module with [terraform-aws-transit-gateway module](#) which handles the creation of TGW resources and VPC attachments. See [complete example there](#).

VPC CIDR from AWS IP Address Manager (IPAM)

It is possible to have your VPC CIDR assigned from an [AWS IPAM Pool](#). However, In order to build subnets within this module Terraform must know subnet CIDRs to properly plan the amount of resources to build. Since CIDR is derived by IPAM by calling CreateVpc this is not possible within a module unless cidr is known ahead of time. You can get around this by "previewing" the CIDR and then using that as the subnet values.

Note: Due to race conditions with `terraform plan`, it is not possible to use `ipv4_netmask_length` or a pools `allocation_default_netmask_length` within this module. You must explicitly set the CIDRs for a pool to use.

```
# Find the pool RAM shared to your account
# Info on RAM sharing pools: https://docs.aws.amazon.com/vpc/latest/ipam/share-pool-ipam.html
data "aws_vpc_ipam_pool" "ipv4_example" {
  filter {
    name   = "description"
    values = ["*mypool*"]
  }
}
```

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

```

    values = ["ipv4"]
  }
}

# Preview next CIDR from pool
data "aws_vpc_ipam_preview_next_cidr" "previewed_cidr" {
  ipam_pool_id   = data.aws_vpc_ipam_pool.ipv4_example.id
  netmask_length = 24
}

data "aws_region" "current" {}

# Calculate subnet cidrs from previewed IPAM CIDR
locals {
  partition      = cidrsubnets(data.aws_vpc_ipam_preview_next_cidr.previewed_cidr.cidr, 2, 2)
  private_subnets = cidrsubnets(local.partition[0], 2, 2)
  public_subnets  = cidrsubnets(local.partition[1], 2, 2)
  azs              = formatlist("${data.aws_region.current.name}%s", ["a", "b"])
}

module "vpc_cidr_from_ipam" {
  source      = "terraform-aws-modules/vpc/aws"
  name        = "vpc-cidr-from-ipam"
  ipv4_ipam_pool_id = data.aws_vpc_ipam_pool.ipv4_example.id
  azs         = local.azs
  cidr        = data.aws_vpc_ipam_preview_next_cidr.previewed_cidr.cidr
  private_subnets = local.private_subnets
  public_subnets  = local.public_subnets
}

```

Examples

- [Complete VPC with VPC Endpoints.](#)
- [VPC using IPAM](#)
- [Dualstack IPv4/IPv6 VPC](#)

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

- [Network ACL](#)
- [VPC with Outpost](#)
- [VPC with secondary CIDR blocks](#)
- [VPC with unique route tables](#)
- [Simple VPC](#)
- [VPC Flow Logs](#)
- [Few tests and edge case examples](#)

Contributing

Report issues/questions/feature requests on in the [issues](#) section.

Full contributing [guidelines](#) are covered [here](#).

Requirements

Name	Version
terraform	>= 1.0
aws	>= 5.0

Providers

Name	Version
aws	>= 5.0

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

No modules.

Resources

Name	Type
aws_cloudwatch_log_group.flow_log	resource
aws_customer_gateway.this	resource
aws_db_subnet_group.database	resource
aws_default_network_acl.this	resource
aws_default_route_table.default	resource
aws_default_security_group.this	resource
aws_default_vpc.this	resource
aws_egress_only_internet_gateway.this	resource
aws_eip.nat	resource
aws_elasticache_subnet_group.elasticache	resource
aws_flow_log.this	resource
aws_iam_policy.vpc_flow_log_cloudwatch	resource
aws_iam_role.vpc_flow_log_cloudwatch	resource
aws_iam_role_policy_attachment.vpc_flow_log_cloudwatch	resource
aws_internet_gateway.this	resource
aws_nat_gateway.this	resource
aws_network_acl.database	resource
aws_network_acl.elasticache	resource

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our Privacy Policy and Cookie Policy.

Name	Type
aws_network_acl.private	resource
aws_network_acl.public	resource
aws_network_acl.redshift	resource
aws_network_acl_rule.database_inbound	resource
aws_network_acl_rule.database_outbound	resource
aws_network_acl_rule.elasticache_inbound	resource
aws_network_acl_rule.elasticache_outbound	resource
aws_network_acl_rule.intra_inbound	resource
aws_network_acl_rule.intra_outbound	resource
aws_network_acl_rule.outpost_inbound	resource
aws_network_acl_rule.outpost_outbound	resource
aws_network_acl_rule.private_inbound	resource
aws_network_acl_rule.private_outbound	resource
aws_network_acl_rule.public_inbound	resource
aws_network_acl_rule.public_outbound	resource
aws_network_acl_rule.redshift_inbound	resource
aws_network_acl_rule.redshift_outbound	resource
aws_redshift_subnet_group.redshift	resource
aws_route.database_dns64_nat_gateway	resource
aws_route.database_internet_gateway	resource
aws_route.database_ipv6_egress	resource
aws_route.database_nat_gateway	resource

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our Privacy Policy and Cookie Policy.

Name	Type
aws_route.private_nat_gateway	resource
aws_route.public_internet_gateway	resource
aws_route.public_internet_gateway_ipv6	resource
aws_route_table.database	resource
aws_route_table.elasticache	resource
aws_route_table.intra	resource
aws_route_table.private	resource
aws_route_table.public	resource
aws_route_table.redshift	resource
aws_route_table_association.database	resource
aws_route_table_association.elasticache	resource
aws_route_table_association.intra	resource
aws_route_table_association.outpost	resource
aws_route_table_association.private	resource
aws_route_table_association.public	resource
aws_route_table_association.redshift	resource
aws_route_table_association.redshift_public	resource
aws_subnet.database	resource
aws_subnet.elasticache	resource
aws_subnet.intra	resource
aws_subnet.outpost	resource
aws_subnet.private	resource

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our Privacy Policy and Cookie Policy.

Name	Type
aws_vpc.this	resource
aws_vpc_dhcp_options.this	resource
aws_vpc_dhcp_options_association.this	resource
aws_vpc_ipv4_cidr_block_association.this	resource
aws_vpn_gateway.this	resource
aws_vpn_gateway_attachment.this	resource
aws_vpn_gateway_route_propagation.intra	resource
aws_vpn_gateway_route_propagation.private	resource
aws_vpn_gateway_route_propagation.public	resource
aws_iam_policy_document.flow_log_cloudwatch_assume_role	data source
aws_iam_policy_document.vpc_flow_log_cloudwatch	data source

Inputs

Name	Description
amazon_side_asn	The Autonomous System Number (ASN) for the Amazon side of the gateway. By default the virt private gateway is created with the current default Amazon ASN
azs	A list of availability zones names or ids in the region
cidr	(Optional) The IPv4 CIDR block for the VPC. CIDR can be explicitly set or it can be derived from using <code>ipv4_netmask_length</code> & <code>ipv4_ipam_pool_id</code>
create_database_internet_gateway_route	Controls if an internet gateway route for public database access should be created
create_database_nat_gateway_route	Controls if a nat gateway route should be created to give internet access to the database subne
create_database_subnet_group	Controls if database subnet group should be created (n.b. database_subnets must also be set)

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
<code>create_elasticache_subnet_group</code>	Controls if elasticache subnet group should be created
<code>create_elasticache_subnet_route_table</code>	Controls if separate route table for elasticache should be created
<code>create_flow_log_cloudwatch_iam_role</code>	Whether to create IAM role for VPC Flow Logs
<code>create_flow_log_cloudwatch_log_group</code>	Whether to create CloudWatch log group for VPC Flow Logs
<code>create_igw</code>	Controls if an Internet Gateway is created for public subnets and the related routes that connect
<code>create_redshift_subnet_group</code>	Controls if redshift subnet group should be created
<code>create_redshift_subnet_route_table</code>	Controls if separate route table for redshift should be created
<code>create_vpc</code>	Controls if VPC should be created (it affects almost all resources)
<code>customer_gateway_tags</code>	Additional tags for the Customer Gateway
<code>customer_gateways</code>	Maps of Customer Gateway's attributes (BGP ASN and Gateway's Internet-routable external IP address)
<code>customer_owned_ipv4_pool</code>	The customer owned IPv4 address pool. Typically used with the <code>map_customer_owned_ip_on_launch</code> argument. The <code>outpost_arn</code> argument must be specified when configured
<code>database_acl_tags</code>	Additional tags for the database subnets network ACL
<code>database_dedicated_network_acl</code>	Whether to use dedicated network ACL (not default) and custom rules for database subnets
<code>database_inbound_acl_rules</code>	Database subnets inbound network ACL rules

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
database_route_table_tags	Additional tags for the database route tables
database_subnet_assign_ipv6_address_on_creation	Specify true to indicate that network interfaces created in the specified subnet should be assign IPv6 address. Default is <code>false</code>
database_subnet_enable_dns64	Indicates whether DNS queries made to the Amazon-provided DNS Resolver in this subnet should return synthetic IPv6 addresses for IPv4-only destinations. Default: <code>true</code>
database_subnet_enable_resource_name_dns_a_record_on_launch	Indicates whether to respond to DNS queries for instance hostnames with DNS A records. Default: <code>false</code>
database_subnet_enable_resource_name_dns_aaaa_record_on_launch	Indicates whether to respond to DNS queries for instance hostnames with DNS AAAA records. Default: <code>true</code>
database_subnet_group_name	Name of database subnet group
database_subnet_group_tags	Additional tags for the database subnet group
database_subnet_ipv6_native	Indicates whether to create an IPv6-only subnet. Default: <code>false</code>
database_subnet_ipv6_prefixes	Assigns IPv6 database subnet id based on the Amazon provided /56 prefix base 10 integer (0-255). Must be of equal length to the corresponding IPv4 subnet list
database_subnet_names	Explicit values to use in the Name tag on database subnets. If empty, Name tags are generated
database_subnet_private_dns_hostname_type_on_launch	The type of hostnames to assign to instances in the subnet at launch. For IPv6-only subnets, an instance DNS name must be based on the instance ID. For dual-stack and IPv4-only subnets, you specify whether DNS names use the instance IPv4 address or the instance ID. Valid values: <code>ip-n</code> <code>resource-name</code>

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
database_subnets	A list of database subnets inside the VPC
default_network_acl_egress	List of maps of egress rules to set on the Default Network ACL
default_network_acl_ingress	List of maps of ingress rules to set on the Default Network ACL

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
default_network_acl_name	Name to be used on the Default Network ACL
default_network_acl_tags	Additional tags for the Default Network ACL
default_route_table_name	Name to be used on the default route table
default_route_table_propagating_vgws	List of virtual gateways for propagation
default_route_table_routes	Configuration block of routes. See https://registry.terraform.io/providers/hashicorp/aws/latest/docs/resources/default_route_table
default_route_table_tags	Additional tags for the default route table
default_security_group_egress	List of maps of egress rules to set on the default security group
default_security_group_ingress	List of maps of ingress rules to set on the default security group
default_security_group_name	Name to be used on the default security group
default_security_group_tags	Additional tags for the default security group
default_vpc_enable_dns_hostnames	Should be true to enable DNS hostnames in the Default VPC
default_vpc_enable_dns_support	Should be true to enable DNS support in the Default VPC
default_vpc_name	Name to be used on the Default VPC
default_vpc_tags	Additional tags for the Default VPC
dhcp_options_domain_name	Specifies DNS name for DHCP options set (requires enable_dhcp_options set to true)
dhcp_options_domain_name_servers	Specify a list of DNS server addresses for DHCP options set, default to AWS provided (requires enable_dhcp_options set to true)
dhcp_options_netbios_name_servers	Specify a list of netbios servers for DHCP options set (requires enable_dhcp_options set to true)

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
dhcp_options_tags	Additional tags for the DHCP option set (requires enable_dhcp_options set to true)
elasticache_acl_tags	Additional tags for the elasticache subnets network ACL
elasticache_dedicated_network_acl	Whether to use dedicated network ACL (not default) and custom rules for elasticache subnets
elasticache_inbound_acl_rules	Elasticache subnets inbound network ACL rules
elasticache_outbound_acl_rules	Elasticache subnets outbound network ACL rules
elasticache_route_table_tags	Additional tags for the elasticache route tables
elasticache_subnet_assign_ipv6_address_on_creation	Specify true to indicate that network interfaces created in the specified subnet should be assign IPv6 address. Default is <code>false</code>
elasticache_subnet_enable_dns64	Indicates whether DNS queries made to the Amazon-provided DNS Resolver in this subnet should

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
	false
elasticache_subnet_enable_resource_name_dns_aaaa_record_on_launch	Indicates whether to respond to DNS queries for instance hostnames with DNS AAAA records. Default: true
elasticache_subnet_group_name	Name of elasticache subnet group
elasticache_subnet_group_tags	Additional tags for the elasticache subnet group
elasticache_subnet_ipv6_native	Indicates whether to create an IPv6-only subnet. Default: false
elasticache_subnet_ipv6_prefixes	Assigns IPv6 elasticache subnet id based on the Amazon provided /56 prefix base 10 integer (0- Must be of equal length to the corresponding IPv4 subnet list
elasticache_subnet_names	Explicit values to use in the Name tag on elasticache subnets. If empty, Name tags are generate
elasticache_subnet_private_dns_hostname_type_on_launch	The type of hostnames to assign to instances in the subnet at launch. For IPv6-only subnets, an instance DNS name must be based on the instance ID. For dual-stack and IPv4-only subnets, yo specify whether DNS names use the instance IPv4 address or the instance ID. Valid values: ip-n resource-name
elasticache_subnet_suffix	Suffix to append to elasticache subnets name
elasticache_subnet_tags	Additional tags for the elasticache subnets
elasticache_subnets	A list of elasticache subnets inside the VPC
enable_dhcp_options	Should be true if you want to specify a DHCP options set with a custom domain name, DNS sen NTP servers, netbios servers, and/or netbios server type
enable_dns_hostnames	Should be true to enable DNS hostnames in the VPC
enable_dns_support	Should be true to enable DNS support in the VPC
enable_flow_log	Whether or not to enable VPC Flow Logs
enable_ipv6	Requests an Amazon-provided IPv6 CIDR block with a /56 prefix length for the VPC. You cannot specify the range of IP addresses, or the size of the CIDR block
enable_nat_gateway	Should be true if you want to provision NAT Gateways for each of your private networks
enable_network_address_usage_metrics	Determines whether network address usage metrics are enabled for the VPC

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
external_nat_ip_ids	List of EIP IDs to be assigned to the NAT Gateways (used in combination with reuse_nat_ips)
external_nat_ips	List of EIPs to be used for <code>nat_public_ips</code> output (used in combination with reuse_nat_ips and external_nat_ip_ids)
flow_log_cloudwatch_iam_role_arn	The ARN for the IAM role that's used to post flow logs to a CloudWatch Logs log group. When flow_log_destination_arn is set to ARN of Cloudwatch Logs, this argument needs to be provided
flow_log_cloudwatch_log_group_kms_key_id	The ARN of the KMS Key to use when encrypting log data for VPC flow logs
flow_log_cloudwatch_log_group_name_prefix	Specifies the name prefix of CloudWatch Log Group for VPC flow logs
flow_log_cloudwatch_log_group_name_suffix	Specifies the name suffix of CloudWatch Log Group for VPC flow logs
flow_log_cloudwatch_log_group_retention_in_days	Specifies the number of days you want to retain log events in the specified log group for VPC flow logs
flow_log_destination_arn	The ARN of the CloudWatch log group or S3 bucket where VPC Flow Logs will be pushed. If this is a S3 bucket the appropriate permissions need to be set on that bucket's policy. When create_flow_log_cloudwatch_log_group is set to false this argument must be provided
flow_log_destination_type	Type of flow log destination. Can be s3 or cloud-watch-logs
flow_log_file_format	(Optional) The format for the flow log. Valid values: <code>plain-text</code> , <code>parquet</code>
flow_log_hive_compatible_partitions	(Optional) Indicates whether to use Hive-compatible prefixes for flow logs stored in Amazon S3
flow_log_log_format	The fields to include in the flow log record, in the order in which they should appear
flow_log_max_aggregation_interval	The maximum interval of time during which a flow of packets is captured and aggregated into a log record. Valid Values: <code>60</code> seconds or <code>600</code> seconds
flow_log_per_hour_partition	(Optional) Indicates whether to partition the flow log per hour. This reduces the cost and response time for queries
flow_log_traffic_type	The type of traffic to capture. Valid values: ACCEPT, REJECT, ALL
igw_tags	Additional tags for the internet gateway
instance_tenancy	A tenancy option for instances launched into the VPC
intra_acl_tags	Additional tags for the intra subnets network ACL

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
intra_inbound_acl_rules	Intra subnets inbound network ACLs
intra_outbound_acl_rules	Intra subnets outbound network ACLs
intra_route_table_tags	Additional tags for the intra route tables
intra_subnet_assign_ipv6_address_on_creation	Specify true to indicate that network interfaces created in the specified subnet should be assign IPv6 address. Default is <code>false</code>
intra_subnet_enable_dns64	Indicates whether DNS queries made to the Amazon-provided DNS Resolver in this subnet should return synthetic IPv6 addresses for IPv4-only destinations. Default: <code>true</code>
intra_subnet_enable_resource_name_dns_a_record_on_launch	Indicates whether to respond to DNS queries for instance hostnames with DNS A records. Default: <code>false</code>

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
intra_subnet_ipv6_prefixes	Assigns IPv6 intra subnet id based on the Amazon provided /56 prefix base 10 integer (0-256). Must be of equal length to the corresponding IPv4 subnet list
intra_subnet_names	Explicit values to use in the Name tag on intra subnets. If empty, Name tags are generated
intra_subnet_private_dns_hostname_type_on_launch	The type of hostnames to assign to instances in the subnet at launch. For IPv6-only subnets, an instance DNS name must be based on the instance ID. For dual-stack and IPv4-only subnets, you specify whether DNS names use the instance IPv4 address or the instance ID. Valid values: <code>ip-name</code> <code>resource-name</code>
intra_subnet_suffix	Suffix to append to intra subnets name
intra_subnet_tags	Additional tags for the intra subnets
intra_subnets	A list of intra subnets inside the VPC
ipv4_ipam_pool_id	(Optional) The ID of an IPv4 IPAM pool you want to use for allocating this VPC's CIDR
ipv4_netmask_length	(Optional) The netmask length of the IPv4 CIDR you want to allocate to this VPC. Requires specifying a <code>ipv4_ipam_pool_id</code>
ipv6_cidr	(Optional) IPv6 CIDR block to request from an IPAM Pool. Can be set explicitly or derived from <code>ipv6_netmask_length</code> using <code>cidrsubnet</code>
ipv6_cidr_block_network_border_group	By default when an IPv6 CIDR is assigned to a VPC a default <code>ipv6_cidr_block_network_border_group</code> will be set to the region of the VPC. This can be changed to restrict advertisement of public address to specific Network Border Groups such as LocalZones
ipv6_ipam_pool_id	(Optional) IPAM Pool ID for a IPv6 pool. Conflicts with <code>assign_generated_ipv6_cidr_block</code>
ipv6_netmask_length	(Optional) Netmask length to request from IPAM Pool. Conflicts with <code>ipv6_cidr_block</code> . This can be omitted if IPAM pool is set as <code>allocation_default_netmask_length</code> set. Valid values: <code>56</code>
manage_default_network_acl	Should be true to adopt and manage Default Network ACL
manage_default_route_table	Should be true to manage default route table
manage_default_security_group	Should be true to adopt and manage default security group
manage_default_vpc	Should be true to adopt and manage Default VPC
manage_customer_owned_ip_on_launch	Specify true to indicate that network interfaces created in the subnet should be assigned a customer-owned IP address

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
map_public_ip_on_launch	Specify true to indicate that instances launched into the subnet should be assigned a public IP address. Default is <code>false</code>
name	Name to be used on all the resources as identifier
nat_eip_tags	Additional tags for the NAT EIP
nat_gateway_destination_cidr_block	Used to pass a custom destination route for private NAT Gateway. If not specified, the default 0.0.0.0/0 is used as a destination route
nat_gateway_tags	Additional tags for the NAT gateways
one_nat_gateway_per_az	Should be true if you want only one NAT Gateway per availability zone. Requires <code>var.azs</code> to be and the number of <code>public_subnets</code> created to be greater than or equal to the number of availability zones specified in <code>var.azs</code>
outpost_acl_tags	Additional tags for the outpost subnets network ACL
outpost_arn	ARN of Outpost you want to create a subnet in
outpost_az	AZ where Outpost is anchored
outpost_dedicated_network_acl	Whether to use dedicated network ACL (not default) and custom rules for outpost subnets
outpost_inbound_acl_rules	Outpost subnets inbound network ACLs
outpost_outbound_acl_rules	Outpost subnets outbound network ACLs

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
outpost_subnet_assign_ipv6_address_on_creation	Specify true to indicate that network interfaces created in the specified subnet should be assign IPv6 address. Default is <code>false</code>
outpost_subnet_enable_dns64	Indicates whether DNS queries made to the Amazon-provided DNS Resolver in this subnet should return synthetic IPv6 addresses for IPv4-only destinations. Default: <code>true</code>
outpost_subnet_enable_resource_name_dns_a_record_on_launch	Indicates whether to respond to DNS queries for instance hostnames with DNS A records. Default: <code>false</code>
outpost_subnet_enable_resource_name_dns_aaaa_record_on_launch	Indicates whether to respond to DNS queries for instance hostnames with DNS AAAA records. Default: <code>true</code>
outpost_subnet_ipv6_native	Indicates whether to create an IPv6-only subnet. Default: <code>false</code>
outpost_subnet_ipv6_prefixes	Assigns IPv6 outpost subnet id based on the Amazon provided /56 prefix base 10 integer (0-255). Must be of equal length to the corresponding IPv4 subnet list
outpost_subnet_names	Explicit values to use in the Name tag on outpost subnets. If empty, Name tags are generated
outpost_subnet_private_dns_hostname_type_on_launch	The type of hostnames to assign to instances in the subnet at launch. For IPv6-only subnets, an instance DNS name must be based on the instance ID. For dual-stack and IPv4-only subnets, you specify whether DNS names use the instance IPv4 address or the instance ID. Valid values: <code>ip-n</code> <code>resource-name</code>
outpost_subnet_suffix	Suffix to append to outpost subnets name
outpost_subnet_tags	Additional tags for the outpost subnets
outpost_subnets	A list of outpost subnets inside the VPC
private_acl_tags	Additional tags for the private subnets network ACL
private_dedicated_network_acl	Whether to use dedicated network ACL (not default) and custom rules for private subnets

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
private_outbound_acl_rules	Private subnets outbound network ACLs
private_route_table_tags	Additional tags for the private route tables
private_subnet_assign_ipv6_address_on_creation	Specify true to indicate that network interfaces created in the specified subnet should be assign IPv6 address. Default is <code>false</code>
private_subnet_enable_dns64	Indicates whether DNS queries made to the Amazon-provided DNS Resolver in this subnet should return synthetic IPv6 addresses for IPv4-only destinations. Default: <code>true</code>
private_subnet_enable_resource_name_dns_a_record_on_launch	Indicates whether to respond to DNS queries for instance hostnames with DNS A records. Default: <code>false</code>
private_subnet_enable_resource_name_dns_aaaa_record_on_launch	Indicates whether to respond to DNS queries for instance hostnames with DNS AAAA records. Default: <code>true</code>
private_subnet_ipv6_native	Indicates whether to create an IPv6-only subnet. Default: <code>false</code>

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our Privacy Policy and Cookie Policy.

Name	Description
private_subnet_names	Explicit values to use in the Name tag on private subnets. If empty, Name tags are generated
private_subnet_private_dns_hostname_type_on_launch	The type of hostnames to assign to instances in the subnet at launch. For IPv6-only subnets, an instance DNS name must be based on the instance ID. For dual-stack and IPv4-only subnets, you specify whether DNS names use the instance IPv4 address or the instance ID. Valid values: <code>ip-n</code> <code>resource-name</code>
private_subnet_suffix	Suffix to append to private subnets name
private_subnet_tags	Additional tags for the private subnets
private_subnet_tags_per_az	Additional tags for the private subnets where the primary key is the AZ
private_subnets	A list of private subnets inside the VPC
propagate_intra_route_tables_vgw	Should be true if you want route table propagation
propagate_private_route_tables_vgw	Should be true if you want route table propagation
propagate_public_route_tables_vgw	Should be true if you want route table propagation
public_acl_tags	Additional tags for the public subnets network ACL
public_dedicated_network_acl	Whether to use dedicated network ACL (not default) and custom rules for public subnets
public_inbound_acl_rules	Public subnets inbound network ACLs
public_outbound_acl_rules	Public subnets outbound network ACLs

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
public_route_table_tags	Additional tags for the public route tables
public_subnet_assign_ipv6_address_on_creation	Specify true to indicate that network interfaces created in the specified subnet should be assign IPv6 address. Default is <code>false</code>
public_subnet_enable_dns64	Indicates whether DNS queries made to the Amazon-provided DNS Resolver in this subnet should return synthetic IPv6 addresses for IPv4-only destinations. Default: <code>true</code>
public_subnet_enable_resource_name_dns_a_record_on_launch	Indicates whether to respond to DNS queries for instance hostnames with DNS A records. Default: <code>false</code>
public_subnet_enable_resource_name_dns_aaaa_record_on_launch	Indicates whether to respond to DNS queries for instance hostnames with DNS AAAA records. Default: <code>true</code>
public_subnet_ipv6_native	Indicates whether to create an IPv6-only subnet. Default: <code>false</code>
public_subnet_ipv6_prefixes	Assigns IPv6 public subnet id based on the Amazon provided /56 prefix base 10 integer (0-256) be of equal length to the corresponding IPv4 subnet list
public_subnet_names	Explicit values to use in the Name tag on public subnets. If empty, Name tags are generated
public_subnet_private_dns_hostname_type_on_launch	The type of hostnames to assign to instances in the subnet at launch. For IPv6-only subnets, an instance DNS name must be based on the instance ID. For dual-stack and IPv4-only subnets, you specify whether DNS names use the instance IPv4 address or the instance ID. Valid values: <code>ip-n</code> <code>resource-name</code>
public_subnet_suffix	Suffix to append to public subnets name
public_subnet_tags	Additional tags for the public subnets
public_subnet_tags_per_az	Additional tags for the public subnets where the primary key is the AZ
public_subnets	A list of public subnets inside the VPC

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
redshift_acl_tags	Additional tags for the redshift subnets network ACL
redshift_dedicated_network_acl	Whether to use dedicated network ACL (not default) and custom rules for redshift subnets
redshift_inbound_acl_rules	Redshift subnets inbound network ACL rules
redshift_outbound_acl_rules	Redshift subnets outbound network ACL rules
redshift_route_table_tags	Additional tags for the redshift route tables
redshift_subnet_assign_ipv6_address_on_creation	Specify true to indicate that network interfaces created in the specified subnet should be assign IPv6 address. Default is <code>false</code>
redshift_subnet_enable_dns64	Indicates whether DNS queries made to the Amazon-provided DNS Resolver in this subnet should return synthetic IPv6 addresses for IPv4-only destinations. Default: <code>true</code>

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our Privacy Policy and Cookie Policy.

Name	Description
redshift_subnet_enable_resource_name_dns_aaaa_record_on_launch	Indicates whether to respond to DNS queries for instance hostnames with DNS AAAA records. Default: <code>true</code>
redshift_subnet_group_name	Name of redshift subnet group
redshift_subnet_group_tags	Additional tags for the redshift subnet group
redshift_subnet_ipv6_native	Indicates whether to create an IPv6-only subnet. Default: <code>false</code>
redshift_subnet_ipv6_prefixes	Assigns IPv6 redshift subnet id based on the Amazon provided /56 prefix base 10 integer (0-256). Must be of equal length to the corresponding IPv4 subnet list
redshift_subnet_names	Explicit values to use in the Name tag on redshift subnets. If empty, Name tags are generated
redshift_subnet_private_dns_hostname_type_on_launch	The type of hostnames to assign to instances in the subnet at launch. For IPv6-only subnets, an instance DNS name must be based on the instance ID. For dual-stack and IPv4-only subnets, you specify whether DNS names use the instance IPv4 address or the instance ID. Valid values: <code>ip-name</code> , <code>resource-name</code>
redshift_subnet_suffix	Suffix to append to redshift subnets name
redshift_subnet_tags	Additional tags for the redshift subnets
redshift_subnets	A list of redshift subnets inside the VPC
reuse_nat_ips	Should be true if you don't want EIPs to be created for your NAT Gateways and will instead pass in via the 'external_nat_ip_ids' variable
secondary_cidr_blocks	List of secondary CIDR blocks to associate with the VPC to extend the IP Address pool
single_nat_gateway	Should be true if you want to provision a single shared NAT Gateway across all of your private networks
tags	A map of tags to add to all resources
use_ipam_pool	Determines whether IPAM pool is used for CIDR allocation
vpc_flow_log_permissions_boundary	The ARN of the Permissions Boundary for the VPC Flow Log IAM Role
vpc_flow_log_tags	Additional tags for the VPC Flow Logs
vpc_tags	Additional tags for the VPC

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
vpn_gateway_tags	Additional tags for the VPN gateway

Outputs

Name	Description
azs	A list of availability zones specified as argument to this module
cgw_arns	List of ARNs of Customer Gateway
cgw_ids	List of IDs of Customer Gateway
database_internet_gateway_route_id	ID of the database internet gateway route
database_ipv6_egress_route_id	ID of the database IPv6 egress route
database_nat_gateway_route_ids	List of IDs of the database nat gateway route
database_network_acl_arn	ARN of the database network ACL
database_network_acl_id	ID of the database network ACL
database_route_table_association_ids	List of IDs of the database route table association
database_route_table_ids	List of IDs of database route tables
database_subnet_arns	List of ARNs of database subnets
database_subnet_group	ID of database subnet group
database_subnet_group_name	Name of database subnet group
database_subnets	List of IDs of database subnets
database_subnets_cidr_blocks	List of cidr_blocks of database subnets
database_subnets_ipv6_cidr_blocks	List of IPv6 cidr_blocks of database subnets in an IPv6 enabled VPC
default_network_acl_id	The ID of the default network ACL

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
default_vpc_arn	The ARN of the Default VPC
default_vpc_cidr_block	The CIDR block of the Default VPC
default_vpc_default_network_acl_id	The ID of the default network ACL of the Default VPC
default_vpc_default_route_table_id	The ID of the default route table of the Default VPC
default_vpc_default_security_group_id	The ID of the security group created by default on Default VPC creation
default_vpc_enable_dns_hostnames	Whether or not the Default VPC has DNS hostname support
default_vpc_enable_dns_support	Whether or not the Default VPC has DNS support
default_vpc_id	The ID of the Default VPC
default_vpc_instance_tenancy	Tenancy of instances spin up within Default VPC
default_vpc_main_route_table_id	The ID of the main route table associated with the Default VPC
dhcp_options_id	The ID of the DHCP options
egress_only_internet_gateway_id	The ID of the egress only Internet Gateway
elasticache_network_acl_arn	ARN of the elasticache network ACL
elasticache_network_acl_id	ID of the elasticache network ACL
elasticache_route_table_association_ids	List of IDs of the elasticache route table association
elasticache_route_table_ids	List of IDs of elasticache route tables
elasticache_subnet_arns	List of ARNs of elasticache subnets
elasticache_subnet_group	ID of elasticache subnet group
elasticache_subnet_group_name	Name of elasticache subnet group
elasticache_subnets	List of IDs of elasticache subnets
elasticache_subnets_cidr_blocks	List of cidr_blocks of elasticache subnets
elasticache_subnets_ipv6_cidr_blocks	List of IPv6 cidr_blocks of elasticache subnets in an IPv6 enabled VPC

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
intra_network_acl_arn	ARN of the intra network ACL
intra_network_acl_id	ID of the intra network ACL
intra_route_table_association_ids	List of IDs of the intra route table association
intra_route_table_ids	List of IDs of intra route tables
intra_subnet_arns	List of ARNs of intra subnets
intra_subnets	List of IDs of intra subnets
intra_subnets_cidr_blocks	List of cidr_blocks of intra subnets
intra_subnets_ipv6_cidr_blocks	List of IPv6 cidr_blocks of intra subnets in an IPv6 enabled VPC
name	The name of the VPC specified as argument to this module
nat_ids	List of allocation ID of Elastic IPs created for AWS NAT Gateway
nat_public_ips	List of public Elastic IPs created for AWS NAT Gateway
natgw_ids	List of NAT Gateway IDs
outpost_network_acl_arn	ARN of the outpost network ACL
outpost_network_acl_id	ID of the outpost network ACL
outpost_subnet_arns	List of ARNs of outpost subnets
outpost_subnets	List of IDs of outpost subnets
outpost_subnets_cidr_blocks	List of cidr_blocks of outpost subnets
outpost_subnets_ipv6_cidr_blocks	List of IPv6 cidr_blocks of outpost subnets in an IPv6 enabled VPC
private_ipv6_egress_route_ids	List of IDs of the ipv6 egress route
private_nat_gateway_route_ids	List of IDs of the private nat gateway route
private_network_acl_arn	ARN of the private network ACL
private_network_acl_id	ID of the private network ACL

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
private_subnet_arns	List of ARNs of private subnets
private_subnets	List of IDs of private subnets
private_subnets_cidr_blocks	List of cidr_blocks of private subnets
private_subnets_ipv6_cidr_blocks	List of IPv6 cidr_blocks of private subnets in an IPv6 enabled VPC
public_internet_gateway_ipv6_route_id	ID of the IPv6 internet gateway route
public_internet_gateway_route_id	ID of the internet gateway route
public_network_acl_arn	ARN of the public network ACL
public_network_acl_id	ID of the public network ACL
public_route_table_association_ids	List of IDs of the public route table association
public_route_table_ids	List of IDs of public route tables
public_subnet_arns	List of ARNs of public subnets
public_subnets	List of IDs of public subnets
public_subnets_cidr_blocks	List of cidr_blocks of public subnets
public_subnets_ipv6_cidr_blocks	List of IPv6 cidr_blocks of public subnets in an IPv6 enabled VPC
redshift_network_acl_arn	ARN of the redshift network ACL
redshift_network_acl_id	ID of the redshift network ACL
redshift_public_route_table_association_ids	List of IDs of the public redshift route table association
redshift_route_table_association_ids	List of IDs of the redshift route table association
redshift_route_table_ids	List of IDs of redshift route tables
redshift_subnet_arns	List of ARNs of redshift subnets
redshift_subnet_group	ID of redshift subnet group
redshift_subnets	List of IDs of redshift subnets

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Name	Description
this_customer_gateway	Map of Customer Gateway attributes
vgw_arn	The ARN of the VPN Gateway
vgw_id	The ID of the VPN Gateway
vpc_arn	The ARN of the VPC
vpc_cidr_block	The CIDR block of the VPC
vpc_enable_dns_hostnames	Whether or not the VPC has DNS hostname support
vpc_enable_dns_support	Whether or not the VPC has DNS support
vpc_flow_log_cloudwatch_iam_role_arn	The ARN of the IAM role used when pushing logs to Cloudwatch log group
vpc_flow_log_destination_arn	The ARN of the destination for VPC Flow Logs
vpc_flow_log_destination_type	The type of the destination for VPC Flow Logs
vpc_flow_log_id	The ID of the Flow Log resource
vpc_id	The ID of the VPC
vpc_instance_tenancy	Tenancy of instances spin up within VPC
vpc_ipv6_association_id	The association ID for the IPv6 CIDR block
vpc_ipv6_cidr_block	The IPv6 CIDR block
vpc_main_route_table_id	The ID of the main route table associated with this VPC
vpc_owner_id	The ID of the AWS account that owns the VPC
vpc_secondary_cidr_blocks	List of secondary CIDR blocks of the VPC

Authors

Module is maintained by [Anton Babenko](#) with help from [these awesome contributors](#).

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).

Apache 2 Licensed. See [LICENSE](#) for full details.

Additional information for users from Russia and Belarus

- Russia has [illegally annexed Crimea in 2014](#) and brought the war in Donbas followed by [full-scale invasion of Ukraine in 2022](#).
 - Russia has brought sorrow and devastations to millions of Ukrainians, killed [thousands of innocent people](#), damaged thousands of buildings including [critical infrastructure](#), caused ecocide by [blowing up a dam](#), [bombed theater](#) in Mariupol that had "Children" marking on the ground, [raped men and boys](#), [deported children](#) in the occupied territoris, and forced millions of people to flee.
 - [Putin khuylo!](#)
-

INTRO



© HashiCorp 2023

LEARN

DOCS

EXTEND

COMMUNITY

STATUS

PRIVACY

SECURITY

TERMS

PRESS KIT

We use cookies and other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#).